Kyaw Soe Lwin

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EDUCATION

University of California San Diego *Jun 2028*
B.S. Data Science
Coursework: Machine Learning, Deep Learning, Computer Vision, Statistical Natural Language Processing

EXPERIENCE

Arcell AI **San Francisco, CA**
AI Engineering Intern *Oct 2026 – Present*
- Engineered an adversarial evaluation harness and synthetic scenario generator for space threat intelligence (space-threat-eval), establishing an 'AI proposes, code verifies' verification architecture.
- Constructed physics-grounded orbital maneuver detection benchmarks with hidden ground-truth manifests, targeting 0.0% false-high-confidence classifications.
- Pinned platform-core submodule architecture, migrated plugin manifest to v0.3.2 schema (30/30 tests passing), and implemented 40-seed astrodynamics stress probes exposing late-burn failure modes.

Data Science Alliance **San Diego, CA**
Data Science Volunteer *Sep 2026 – Present*
- Designed and executed an end-to-end data auditing and validation pipeline across 50K+ geospatial records, raising multi-year report fidelity to 97.5%+ across 380+ downtown city blocks.
- Standardized schemas and built geospatial crosswalks for time-series decomposition and Getis-Ord Gi* hotspot detection to model urban demographic patterns.

PROJECTS

Enterprise Corrective Multimodal RAGOps Platform (GCP) *2026*
- Architected an enterprise multimodal CRAG platform on GCP using Document AI Layout Parser and Vertex AI 1408-d embeddings to index text, Markdown tables, and cropped chart figures.
- Implemented a cyclic LangGraph state machine with dynamic relevance grading and grounded Gemini 2.5 Flash synthesis; evaluated via Ragas CI/CD (0.7519 Faithfulness) on Cloud Run.

LoRA TinyLlama-1.1B Instruction Fine-Tuning *2026*
- Fine-tuned TinyLlama-1.1B with LoRA, adapting only the attention projection layers (q/k/v/o) to cut trainable parameters by over 95% while keeping the base model frozen.
- Applied 4-bit quantization to train on a Colab T4 GPU and resolved a KV-cache/gradient-checkpointing conflict that caused garbled inference output.

Telco Customer Churn Pipeline & Profit Thresholding *2026*
- Built a preprocessing pipeline with data-leakage checks and tuned XGBoost with 30 Optuna trials, achieving 0.844 ROC-AUC and 0.671 PR-AUC on a held-out test set of 1,057 customers.
- Deployed a Streamlit dashboard with SHAP explanations and profit-threshold analysis to recommend outreach cutoffs based on customer lifetime value and contact costs.

HONORS

Jack Kent Cooke Undergraduate Transfer Scholarship
National Semifinalist *2026*
- Selected as one of 485 semifinalists from more than 1,300 applicants nationwide.

SKILLS

Languages: Python, SQL, Java, TypeScript, Bash/Linux
Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LangChain, LangGraph, Hugging Face Transformers
Libraries: Pandas, NumPy, OpenCV, ChromaDB, Optuna, SHAP, Ragas
Tools: AWS, GCP (Vertex AI), Docker, Kubernetes (GKE), PostgreSQL, Git/GitHub, Streamlit, Claude Code
Concepts: Generative AI, LLMs, Multimodal RAG / CRAG, PEFT/LoRA, Quantization (QLoRA), RLHF, Prompt Engineering, Computer Vision, CNNs, Agile/Scrum, OOP, MLOps